Report for Take Home 1:

G06
Chaithanya PS (CS25B018)
Yash Tulshyan (CS25B095)

Circuit for add_sub_16:

Implemented:

- ☑ half_adder
- ☑ full_adder
- ☑ adder16
- ☑ Addsub16
  - ☑ COUT
  - ☑ OVF

Pending:

None

Circuit for shifters:

Implemented:

Implemented shifter using fixed 1,2,4,8 and 16 bit shifters and cascading them based on the set bit in SH[5]

- ☑ left_shift
  - ☑ Maximum Case Handled
- ☑ right_shift
  - ☑ Maximum Case Handled

Pending:

None

Circuit for Comparator

Implemented:

- ☑ cmp_4
- ☑ cmp_16 (signed)

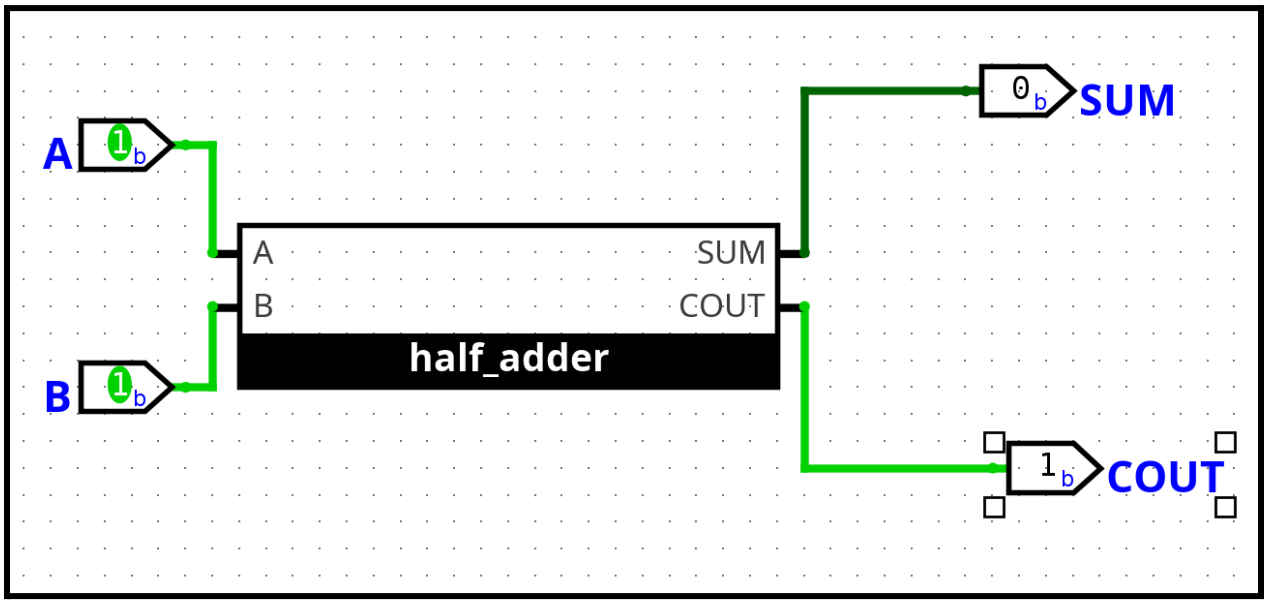
Pending:

None

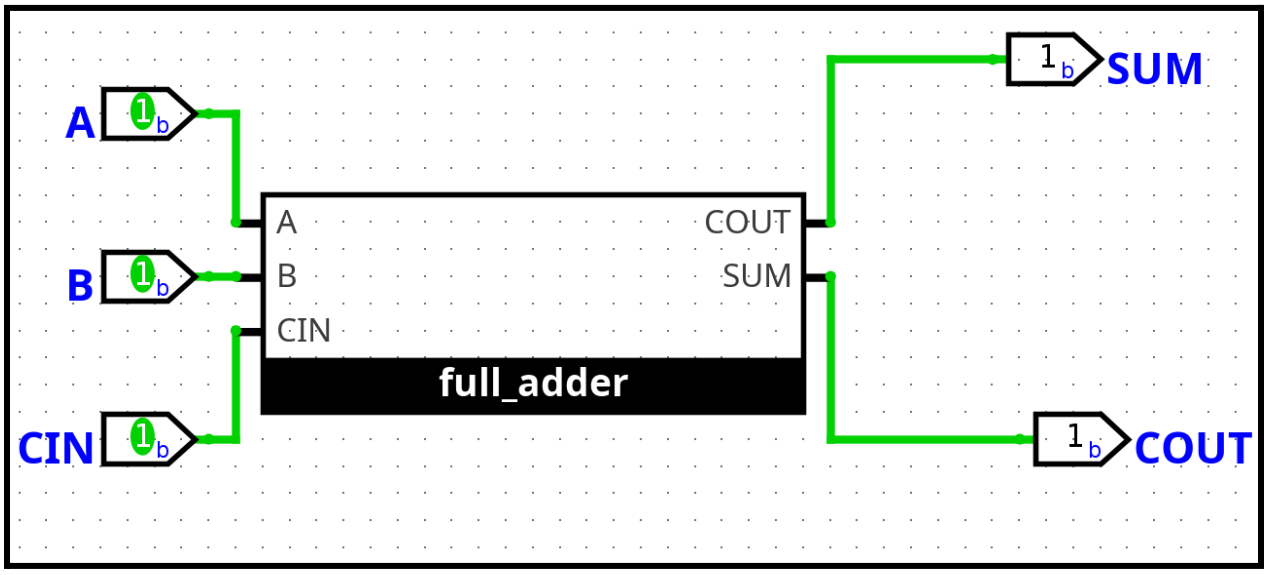
All helper functions saved as file_helper
All main circuits required saved as give in the question

Screenshots:

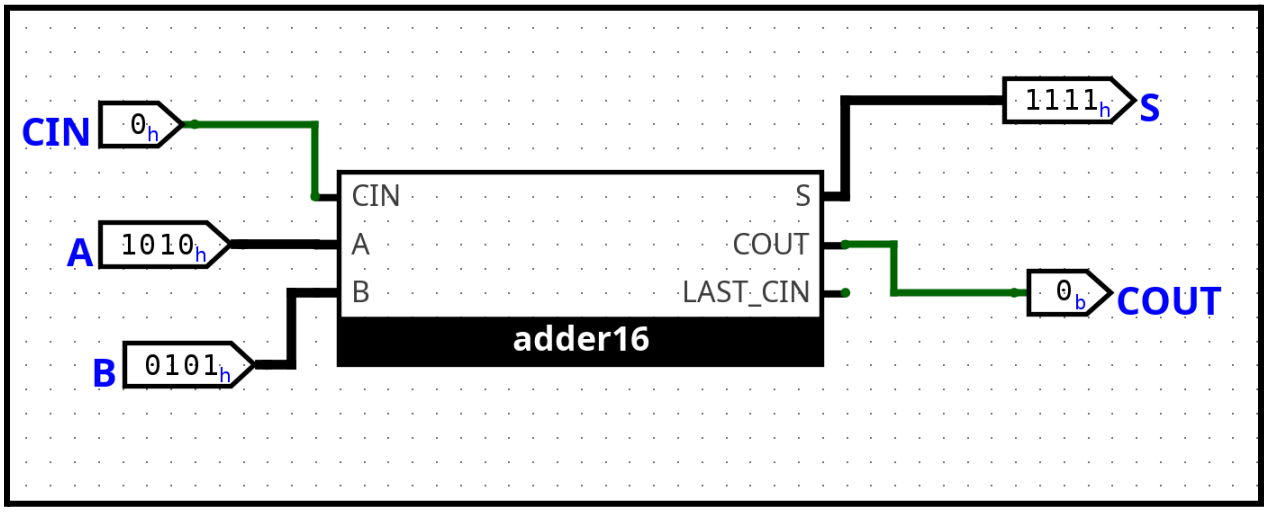
Half adder:



Full adder:

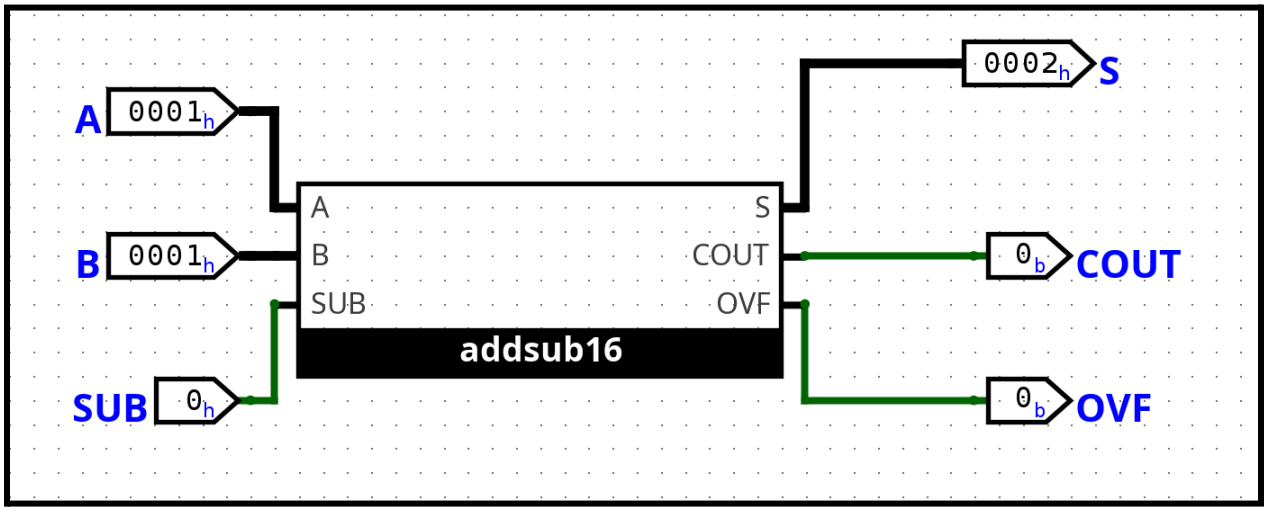


Adder16:

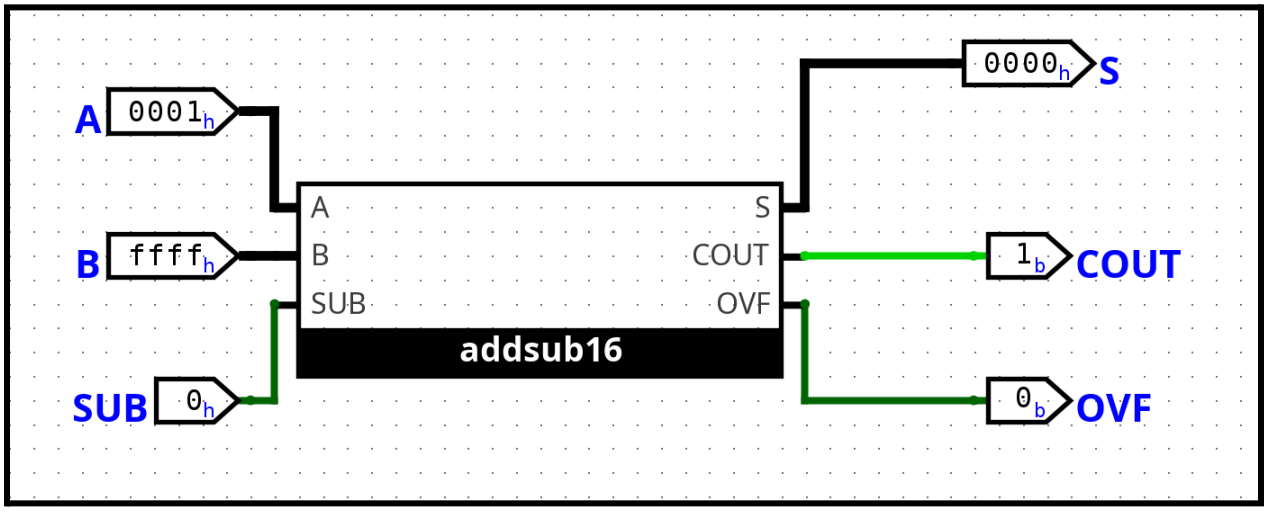


Addsub16:

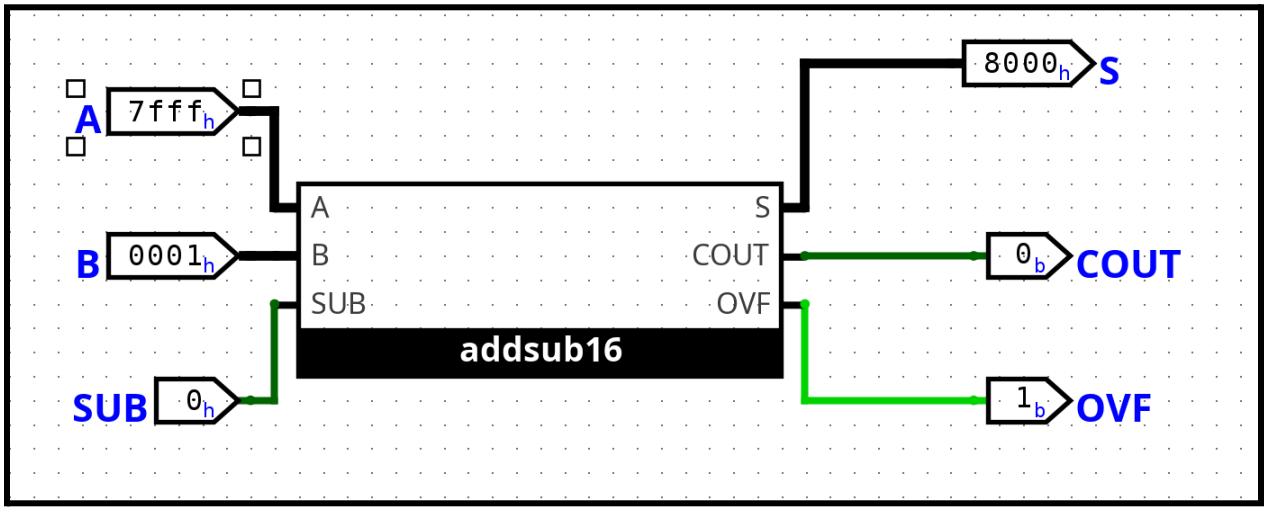
Case 1: COUT = 0, OVF = 0



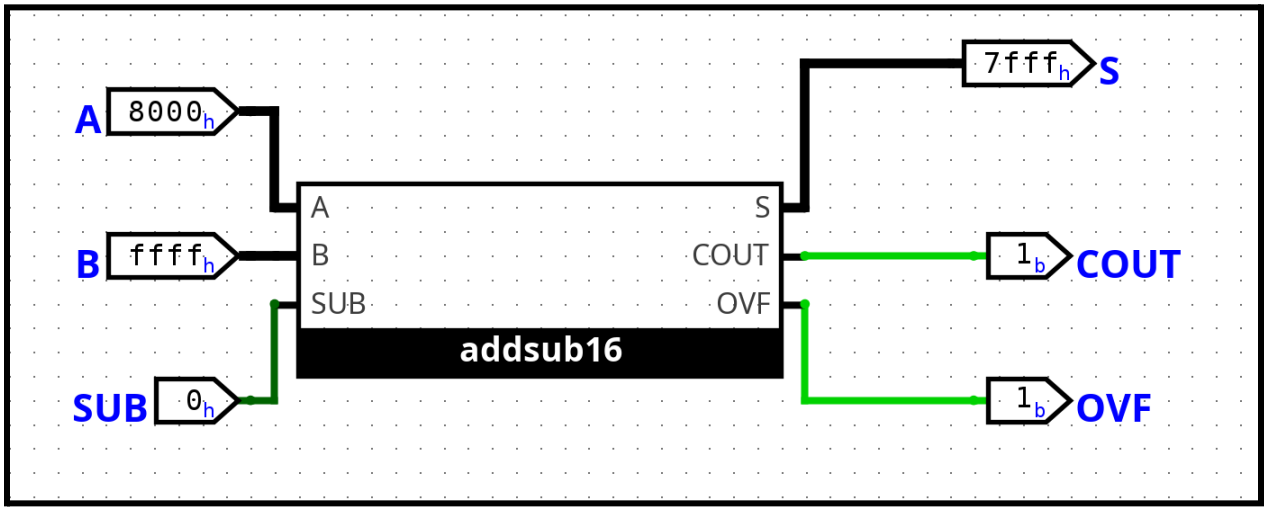
Case 2: COUT = 1, OVF = 0



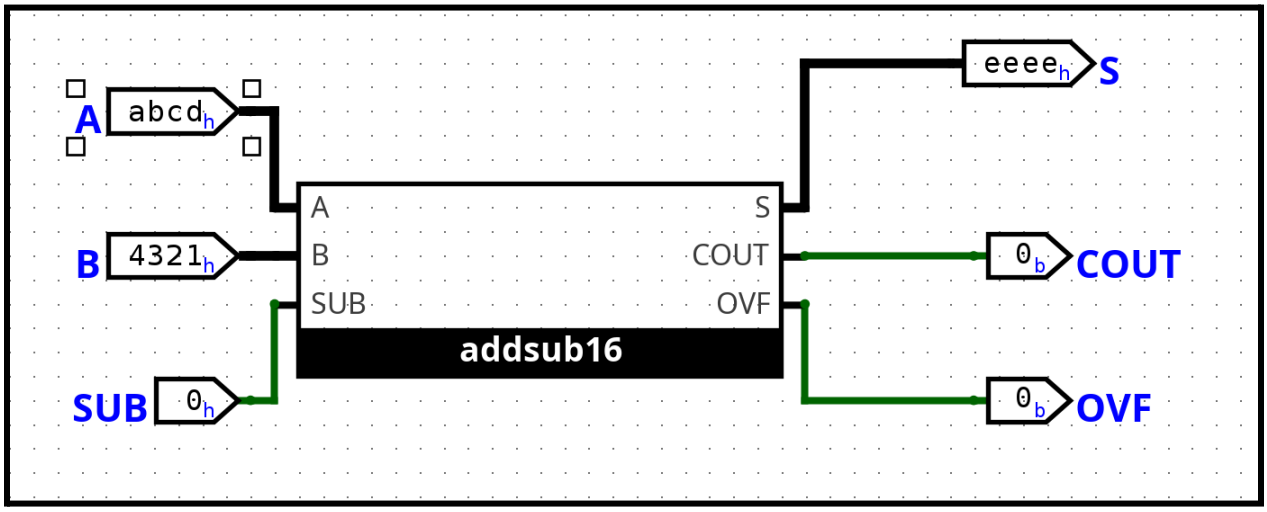
Case 3: COUT = 0, OVF = 1

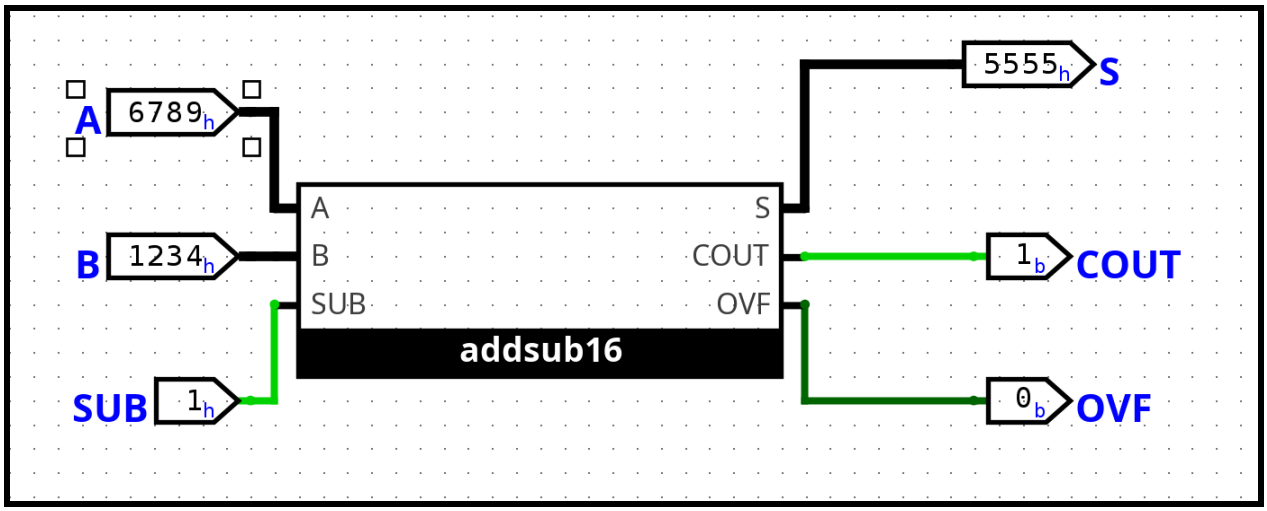


Case 4: COUT = 1, OVF = 1



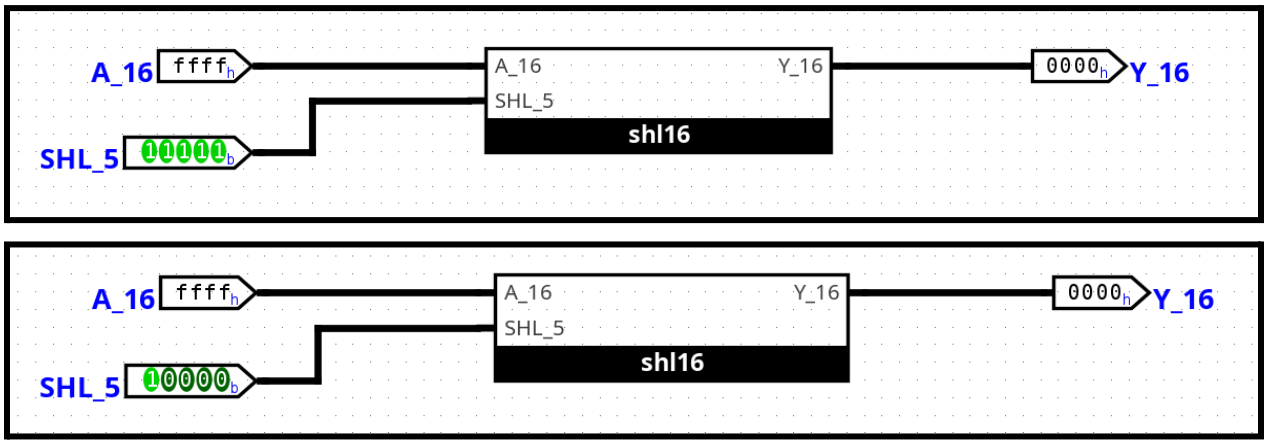
Some random inputs



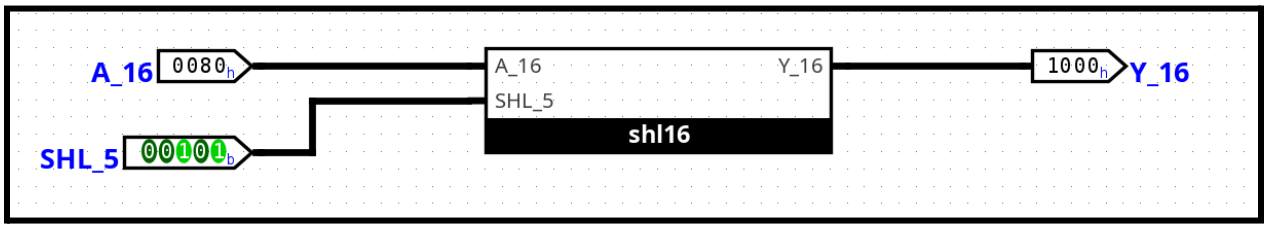

A
6789h
B
1234h
SUB
1h
A
B
SUB
S
COUT
OVF
addsub16
5555h
S
1b
COUT
0b
OVF

Shift Left:

Edge cases: SHL >= 16

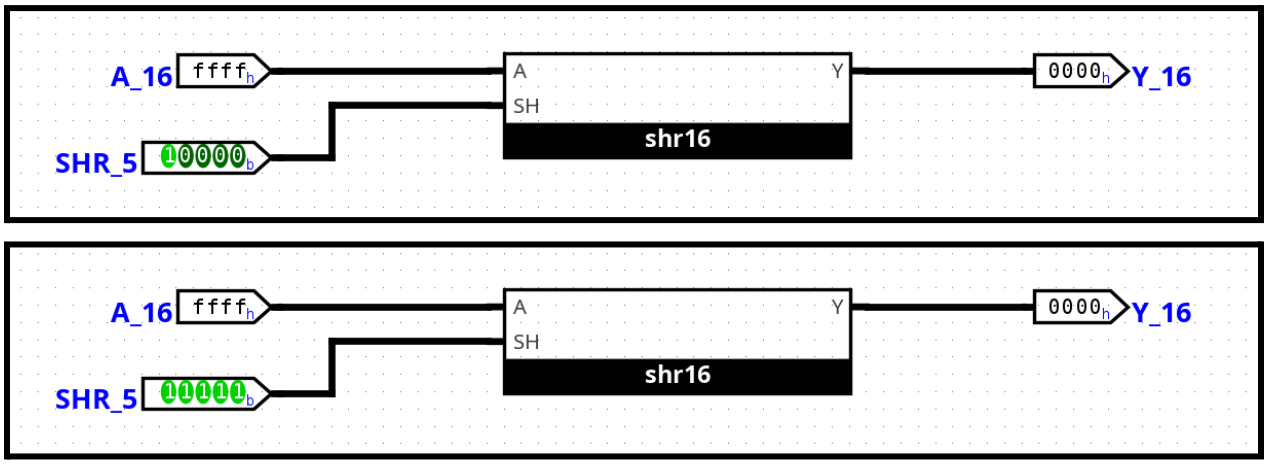


Random Case:

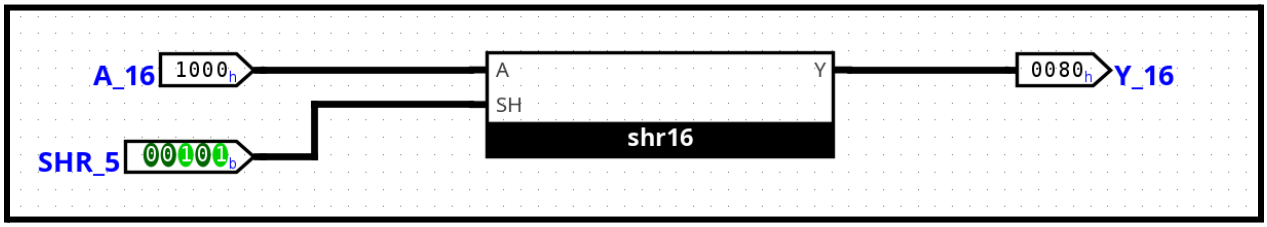


Shift Right:

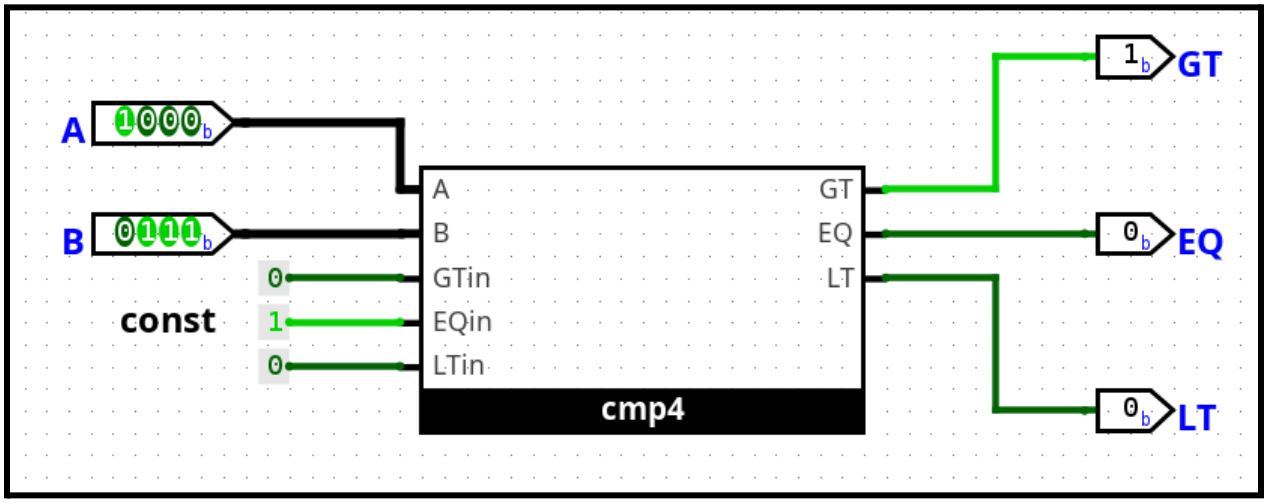
Edge cases: SHR >= 16



Random Case:

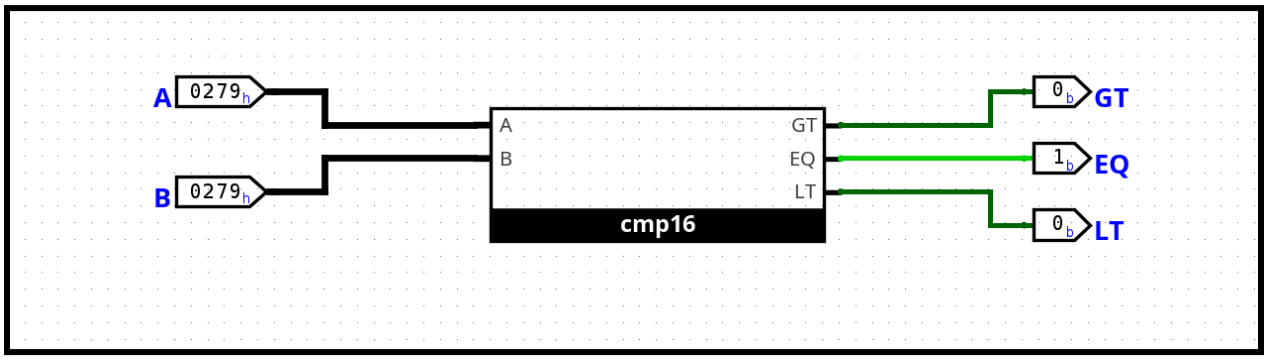


Cmp4:

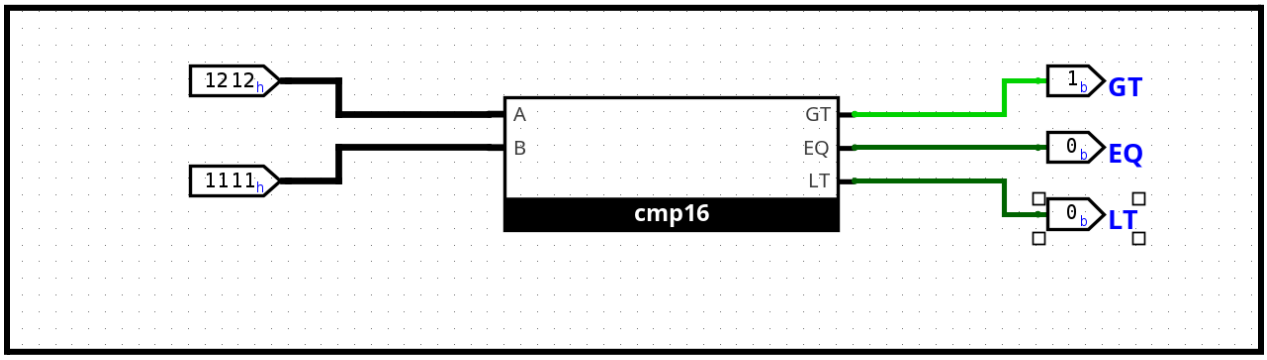


Cmp16 (signed):

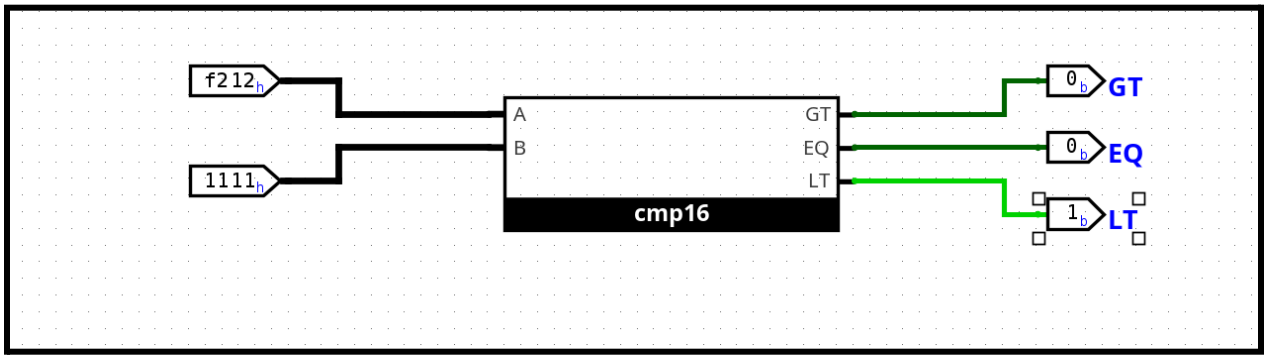
0x0279 = 633



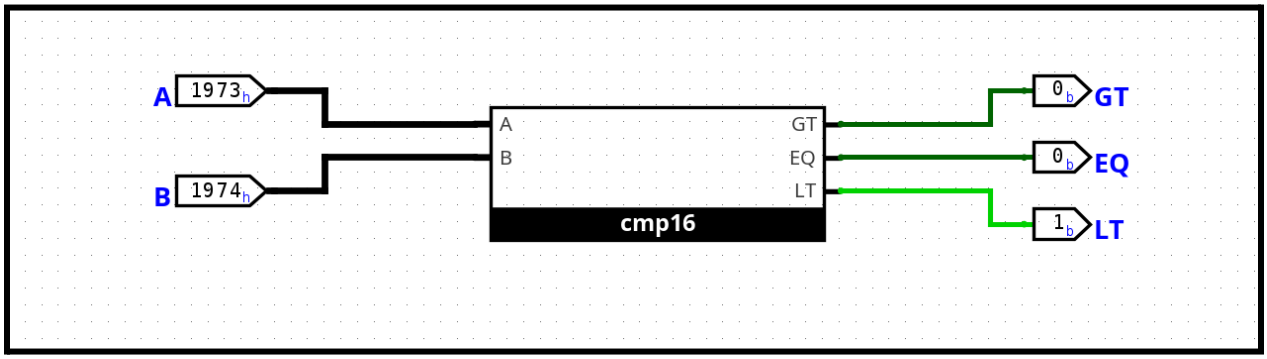
0x1212 = 4626
0x1111 = 4369



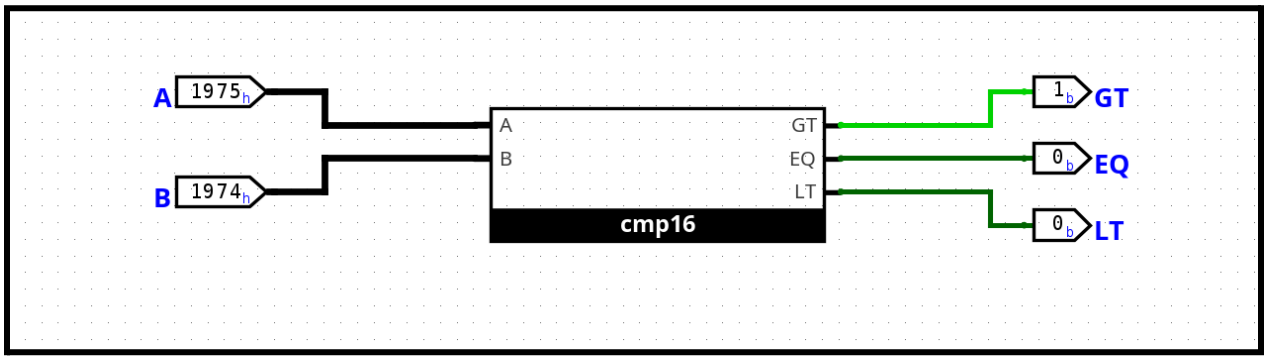
0xf212 = -3566
0x1111 = 4369



0x1973 = 6515
0x1974 = 6516



0x1975 = 6517
0x1974 = 6516



0xffff = -1
0xfeab = -341

